

The Permission Slip, Relocated: Sector Multiplicity, Not the Homogeneity Gap, and the Rescoped J_ω Conjecture

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A companion to Foundation, Revision 19. Short, self-contained.

Revision 7.

1. A dimension mismatch — and what dimension actually settles

[**FACT**] $\text{Aut}(\mathbb{S}) = G_2 \times S_3$ is finite in every sense that matters here: G_2 contributes fourteen continuous dimensions, S_3 a further six discrete elements. The space of normal states on the operational algebra built from $\mathbb{S}$ is infinite-dimensional. A finite-dimensional group's orbits are finite-dimensional, so *transitivity* is excluded outright — no orbit of $\text{Aut}(\mathbb{S})$ can be the whole state space. Dimension-counting stops there. It does not, by itself, exclude *dense* orbits: small groups can have dense orbits on large spaces without contradiction. Claiming density fails “by arithmetic” overstates what arithmetic gives.

[**FACT, cited**] What actually confines the part of $\text{Aut}(\mathbb{S})$ proven to act here is already established, and it is sharper than any dimension count — but it is proven only for the $\mathfrak{so}(3)$ remnant specifically, not for $\text{Aut}(\mathbb{S})$'s image as a whole. The remnant is symplectically implemented, hence exactly χ -conserving by the conservation theorem — confined to a single χ -fiber, never merely close to leaving it — and proven, via the full-strength uniqueness discharge, to fix every family member exactly, not approximately. Whether the rest of G_2 (moving the mediator) or μ (moving the split) acts on this same operational algebra at all is Foundation's own orbit-[OPEN] question, unresolved one level down: those elements presumptively act on differently-constructed, only partially overlapping algebras, and the same-algebra premise this section's confinement claim would need for them is explicitly tagged there as an assumption, not a given.

[**DEF**] The group with a genuine chance of moving broadly around the state space is a different object in kind: $\text{Ad}(U(\mathcal{M}))$, the inner automorphisms of the operational algebra $\mathcal{M}$ itself, implemented by conjugation with its own unitaries — a symmetry of the representation, not of the seed algebra.

2. Connes–Størmer, stated precisely

[**FACT, cited**] For factors of type III_1 , $\text{Ad}(U(\mathcal{M}))$ achieves *homogeneity* of the normal state space (Connes, A. & Størmer, E., “Homogeneity of the state space of factors of type III_1 ,” *J. Funct. Anal.* 28 (1978), 187–196): for any faithful normal states φ, ψ and any $\epsilon > 0$, some unitary u gives $\|\varphi \circ \text{Ad}(u) - \psi\| < \epsilon$. This is what the theorem proves — density of the orbit closure — and it is a statement about *some* automorphism achieving closeness, not a statement that *no* automorphism produces an exact result. Every inner automorphism produces an exact normal state; the theorem is about which target states are reachable exactly, not about exactness failing to occur at all.

[**FACT**] Whether exact equality can fail — whether some ψ is never exactly reached by any $\text{Ad}(u)$ applied to φ — is a separate question the 1978 theorem does not by itself settle, since a dense set can coincide with the whole space. The relevant finer classification (when two normal states are exactly equivalent, and the invariant, the flow of weights, that measures the obstruction) belongs to later, related work (Haagerup, U. & Størmer, E., “Equivalence of normal states on von Neumann algebras and the flow of weights,” *Adv.*

Math. 83 (1990), 180–262). Nothing below rests on the details of that finer classification; it is named here only so that no claim in this document leans on it silently.

[**FACT**] A separate justification sometimes offered alongside this theorem — that type III_1 algebras have no minimal projections and no pure normal states, so no naive collapse-to-pure-outcome picture of measurement applies — does not actually need Connes–Størmer at all. The absence of minimal projections and pure normal states follows from type III alone, at any subtype, with no homogeneity theorem required. Crediting the homogeneity theorem with that particular job is a misattribution worth not repeating, wherever it appears.

3. Why the natural reading of the gap is backwards

[**PROP**] Consider a feature of a state that is *robust* — locally constant, unchanged under sufficiently small perturbations — such as “which of several channels carries the largest expectation value” (the natural technical shape behind an informal description like “which generation is heaviest”). Fix a state φ and a target ψ with a different value of this feature. Local constancy means some neighborhood of ψ , in norm, shares ψ ’s value throughout. Density of φ ’s orbit closure means the orbit meets every nonempty open set — in particular, that neighborhood. So the orbit of φ does not merely approach a state with ψ ’s feature value: by elementary topology, it exactly attains it, via some genuine inner automorphism of φ itself.

[**PROP**] This is the opposite of a permission slip for robust content. Density, applied to any locally-constant feature, forces the orbit to sweep through every value that feature takes anywhere in the state space — meaning inner automorphisms *can and do* move a state exactly between configurations differing in any such feature. What survives as genuinely inaccessible to $\text{Ad}(U(\mathcal{M}))$ is not robust content at all: it is content sensitive to arbitrarily fine distinctions within a single small neighborhood, the opposite of the coarse, stable facts — a heavy generation, an occupied channel, a sign — that would actually be worth protecting. The gap between “arbitrarily close” and “exactly equal” shelters exactly the content physics has no use for, and licenses none of the content this document originally reached for.

[**FACT**] The same structural fact already forced a different construction earlier in this project: χ could not remain a naive state-space distance, and had to be rebuilt as a Bogoliubov-invariant symplectic-spectrum quantity, precisely because ordinary norm-continuous functionals do not survive homogeneity — the identical mechanism at work in §§1–2 above.

4. Where the room actually is

[**FACT, cited**] Foundation already states the genuine source of non-symmetry-forced content, and it needs no theorem-gap to support it. Distinct members of the squeeze-thermal family generate disjoint folia — not approximately, not densely-but-not-quite, but exactly and completely inequivalent representations. Connes–Størmer homogeneity never applies *across* folia at all; it is a statement about the normal states of one factor, and disjoint folia generate different factors from the start. Between sectors, there is no density to erase anything.

[**FACT, cited**] Foundation states further that which member of the family is physically realized is empirical input, not part of the formal starting point. This is the permission slip, already frozen as doctrine: room for genuinely free, non-derived content exists because the formalism identifies a *multiplicity* of exactly inequivalent sectors and explicitly declines to select among them internally — not because any inner-automorphism orbit fails, by an epsilon, to close.

[**INTERPRETATION**] Read this way, §3’s correction and this section’s restatement are the same finding from two directions: the *within*-folium freedom this document originally reached for does not exist, having

been proven not to survive homogeneity; the *between*-folia freedom it should have reached for was already there, stated plainly, needing no rescue.

5. The specific candidate, rescoped

[**FACT**] Conjugation in any Cayley–Dickson algebra fixes the real line and negates the entire imaginary part: for $x \in \text{Im}(\mathbb{S})$, $\bar{x} = -x$, without exception, at every level of the doubling tower. This holds for $\mathbb{S}$ ’s own conjugation regardless of which subspace is later quantized — it is not itself a bosonic-versus-fermionic question.

[**FACT**] What *does* depend on which subspace is quantized is what this conjugation induces on the resulting quantum structure. The current, purely bosonic operational algebra quantizes only $\text{Im}(\text{Fix}(\tau))$; there is no route, within that construction, to see any contribution from the complementary half at all. A comparison to J_ω can only be posed, right now, within that seven-dimensional restriction — and even there, only once an explicit bridge is built: $\mathbb{S}$ ’s conjugation is a map on a sixteen-dimensional algebra, J_ω an antiunitary on a GNS Hilbert space, and no map between them is specified by anything written so far. Until that bridge exists, “aligns or mismatches” is not yet a sentence with a truth value.

[**PROP**] The vacuity concern is resolved, and resolved the way §4 anticipated: negation on a symplectic space preserves the symplectic form exactly ($\sigma(-v, -w) = \sigma(v, w)$, by bilinearity), so it lifts not to something trivial but to a genuine **unitary** — ordinary parity, the operator a total phase-space sign-flip becomes upon quantization, a substantial discrete symmetry in its own right. The candidate is not empty.

[**PROP**] But this resolution exposes a sharper problem than the one it answers. J_ω , from Tomita–Takesaki theory, is **antiunitary** by construction, always. A unitary can never equal an antiunitary — they act oppositely on the imaginary unit, and no nonzero operator satisfies both. So even once the bridge of §4 is built using $\mathbb{S}$ ’s own conjugation directly, the comparison still fails to typecheck: parity, unitary, cannot be compared for equality against J_ω , antiunitary. A well-posed comparison needs J_ω set against some genuinely *antiunitary* composite — parity alone is the wrong kind of object, regardless of what it turns out to equal.

[**FACT**] Producing an antiunitary from this construction needs an antisymplectic map: one reversing the sign of σ , since antisymplectic maps are exactly the ones quantizing to antiunitaries. Two natural candidates exist, and they behave oppositely under the derived flow. Checked directly: $(X, P) \mapsto (X, -P)$ and $(X, P) \mapsto (-X, P)$ are both antisymplectic, and conjugating the derived squeeze flow by either leaves the flow *unchanged* — they commute with it, because each is diagonal in the same basis the flow already is, and diagonal maps commute. The exchange $(X, P) \mapsto (P, X)$ is also antisymplectic, but conjugation by it *inverts* the flow exactly. Reflections commute; the exchange reverses. Which of these families is the right partner for J_ω is settled below by direct appeal to modular theory, not by guessing.

[**PROP**] Tomita–Takesaki theory fixes this outright. The defining relation $J\Delta J = \Delta^{-1}$, together with J ’s antiunitarity, forces $J\Delta^{it}J = \Delta^{it}$ for every real t : conjugating the scalar it by an antiunitary flips its sign to $-it$, and conjugating $\ln \Delta$ by J flips its sign too (since $J\Delta J = \Delta^{-1}$ gives $J(\ln \Delta)J = -\ln \Delta$), so the two sign flips cancel exactly. J therefore *commutes* with every modular unitary Δ^{it} — verified here two ways, by direct operator algebra and by a finite-dimensional numerical example (a genuine non-tracial Tomita–Takesaki pair, J, Δ built from a density matrix with distinct eigenvalues), agreeing to machine precision at several values of t . For a quasi-free KMS state, Δ^{it} implements the second-quantized derived flow, reparametrized by the KMS inverse temperature ($\sigma_t^\omega = S_{-\beta t}$, the standard relation between modular and physical time) — a rescaling irrelevant to which maps commute with it, since commuting with a flow and commuting with any reparametrization of it are the same condition. So J_ω ’s one-particle shadow

commutes with the flow, but commuting alone is not yet enough to isolate the reflections: any map preserving the flow's two eigenlines commutes with it, and antisymplecticity alone permits the whole one-parameter family $(X, P) \mapsto (aX, -P/a)$, of which the reflections are just two points ($a = \pm 1$). What narrows this family to exactly those two points is a second Tomita–Takesaki fact this construction has not yet spent: $J^2 = 1$. Applying $(a, -1/a)$ twice gives $(a^2 X, P/a^2)$, and involutivity forces $a^2 = 1$ — precisely $(X, -P)$ and $(-X, P)$, nothing else. (For modes of distinct frequency, commuting with the flow also forbids mixing between them, so this per-mode analysis is the complete picture; degenerate frequencies would enlarge the candidate set, a spectral dependency folded into the same construction-specific caveat as the mediator-dependence below.) J_ω 's one-particle shadow is accordingly exactly the *flow-commuting, involutive* kind of antisymplectic map — the reflections, not the exchange. The exchange keeps a genuine, separate role: it is this framework's actual time-reversal T , related to J_ω by the CRT-type identity the precedent below describes, not by equality — J_ω is compared against the reflections; T is a different object entirely, sitting one level away from this comparison.

[FACT, proven — resolved by companion note, two independent constructions in agreement] The well-posed form of the comparison is J_ω 's one-particle shadow against the canonical flow-commuting reflections $(X, -P)$ and $(-X, P)$, and the answer is **alignment**: J_ω 's one-particle shadow is exactly $(X, -P)$, verified via two independently-constructed computations — a full Fock-space GNS purification and a direct one-particle Araki–Woods embedding — agreeing exactly, including on the exact (β -independent, not approximate) character of the result and the correct exclusion of the exchange candidate (J_ω Resolved: The One-Particle Shadow Matches $(X, -P)$, companion note). The sign-twin candidate $(-X, P)$ was never independently distinct from $(X, -P)$: as operators, $(-X, P) = -(X, -P)$, so the two-candidate list closes to one, resolved. One dependency still carried explicitly: these reflections are defined relative to the specific X – P splitting this construction supplies, itself built from the mediator direction — so the comparison implicitly consults the same free choice Foundation's own orbit-[OPEN] question tracks, and inherits that question's unresolved status. This result has not yet had a check from outside this collaboration.

[FACT, cited] The general shape of this candidate has real precedent, independent of how the comparison above resolves. In the Bisognano–Wichmann setting, the modular conjugation of a wedge algebra implements a CRT-type reflection on the underlying spacetime structure — the established bridge, in algebraic quantum field theory, between modular theory and discrete spacetime symmetries. J_ω as the natural site for a reflection-type asymmetry is accordingly not a bare hunch; the derivation above shows this project's own construction instantiates the same pattern by the same mechanism, not by analogy.

[SCOPE] The content most relevant to an actual handedness- or generation-tied asymmetry — the kind associated with “chirality” in the wider discussion of this project's honest limits — plausibly requires the full sixteen dimensions and hence the fermionic sector, which does not yet exist. This conjecture accordingly carries the same status as the prospective τ -inclusion elsewhere in this project: contingent on that construction, not evaluable in full ahead of it. The bosonic-level question — whether J_ω 's shadow equals either reflection outright, restricted to the current sector — is a smaller, prior, and separately worthwhile computation.

6. What this document is

A short note establishing where this project's actual room for non-symmetry-forced content lives, and where it does not. It proves nothing about CP violation, chirality, or any specific physical asymmetry.

[LEDGER] Resolved: whether J_ω 's one-particle shadow equals either canonical reflection $(X, -P)$ or $(-X, P)$, restricted to the current bosonic sector — closed, alignment, exact at every temperature

(J_ω Resolved, companion note; pending external check). Still open: the same question’s dependence on the mediator-direction choice, tying it to Foundation’s orbit-[OPEN]; whether the fuller, chirality-relevant version of this comparison is even well-posed before the fermionic sector exists — now better-motivated, given the bosonic sector’s proven cleanliness points any such content toward the fermionic side specifically; and, prior to all of these, whether “CP violation” is the right physical reading of any eventual mismatch at all, as opposed to some other asymmetry, or none. This document inherits every open item of Foundation, Revision 19, without exception, and adds these.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author’s responsibility.